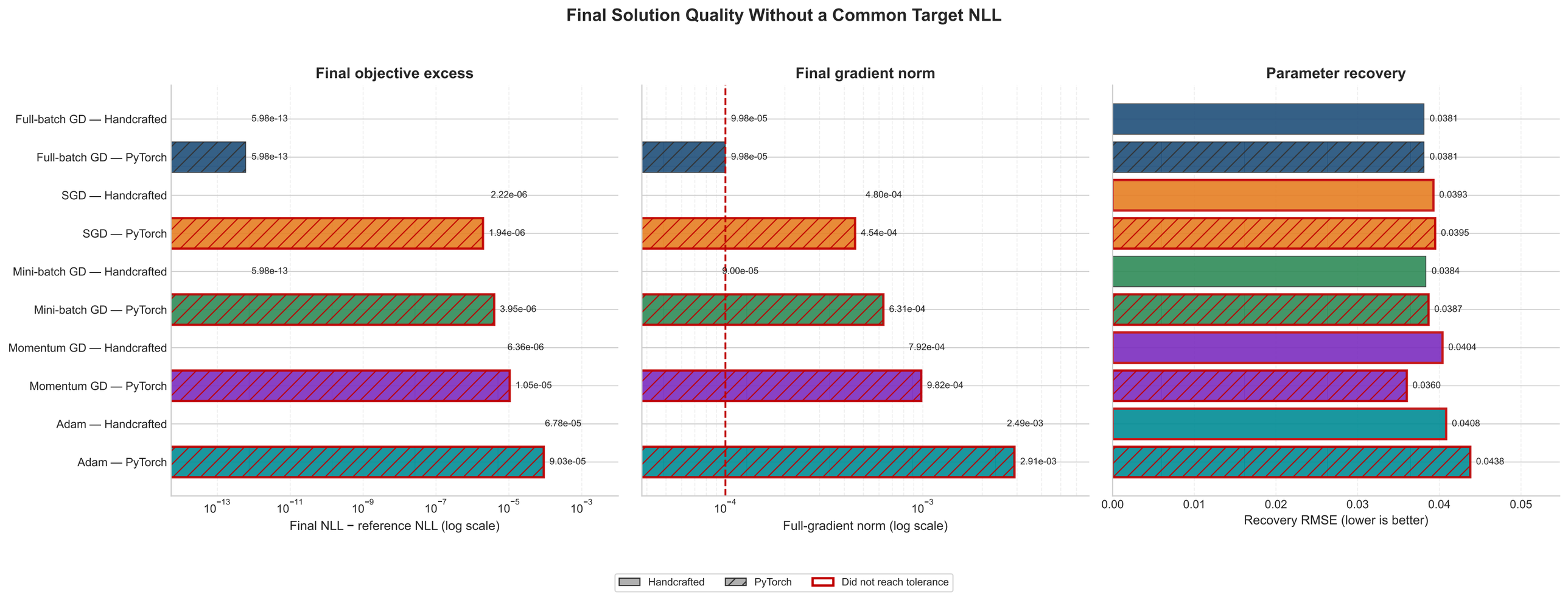
Final Solution Quality Without a Common Target NLL
Final objective excess
Final gradient norm
Parameter recovery
Full-batch GD — Handcrafted
Full-batch GD — PyTorch
SGD — Handcrafted
SGD — PyTorch
Mini-batch GD — Handcrafted
Mini-batch GD — PyTorch
Momentum GD — Handcrafted
Momentum GD — PyTorch
Adam — Handcrafted
Adam — PyTorch
5.98e-13
5.98e-13
2.22e-06
1.94e-06
5.98e-13
3.95e-06
6.36e-06
1.05e-05
6.78e-05
9.03e-05
9.98e-05
9.98e-05
4.80e-04
4.54e-04
9.00e-05
6.31e-04
7.92e-04
9.82e-04
2.49e-03
2.91e-03
0.0381
0.0381
0.0393
0.0395
0.0384
0.0387
0.0404
0.0360
0.0408
0.0438
10^{-13}
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
Final NLL − reference NLL (log scale)
10^{-4}
10^{-3}
Full-gradient norm (log scale)
0.00
0.01
0.02
0.03
0.04
0.05
Recovery RMSE (lower is better)
Handcrafted
PyTorch
Did not reach tolerance